

Lunar Surface Analysis via XRF

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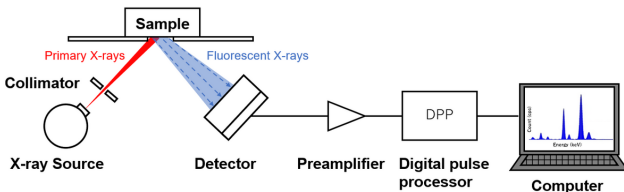
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Introduction

- **Goal:** Identification and quantification of chemical elements on the lunar surface.
- **Challenge:** Remote sensing via X-ray Fluorescence (XRF) produces complex, noisy, and high-dimensional spectral data.
- **Methodology:** Using Machine Learning to automate and refine the interpretation of these spectra.
- **Data Strategy:** Development of a custom dataset and utilization of multiel-spectra for high-fidelity synthetic data generation.

XRF Spectroscopy Fundamentals

- High-energy X-rays/Gamma rays irradiate the sample.
- Secondary X-rays (fluorescence) are emitted, unique to each element.
- The detector captures a spectrum where energy peaks correspond to specific atoms.



The analysis focuses on determining the chemical elements within the XRF spectra through three interconnected computational tasks:

- 1 Classification
- 2 Regression
- 3 Denoising

The classification task focus on identifying the elemental composition by analyzing spectral signatures.

Analytical stages

- **Multi-label classification** across a pool of 41 potential elements.
- **Multi-class classification** across 5 classes defined by the number of elements present in the sample.

Synthetic Data Generation

To train models, we developed a Generator that simulates physical variations in XRF spectra.

- **Composition:** Samples contain **0 to 5** elements randomly selected from **41** possible candidates.
- **Stochasticity:** To ensure model robustness, each spectrum is generated by randomly varying operating conditions and noise levels simulating real-world measurement variability.

Preprocessing Strategies and Architectures

Preprocessing Scaling

Tested to handle high dynamic range in spectral counts:

- **Standard:** 0 mean and 1 variance.
- **MinMax:** Rescaling to $[0, 1]$ range.
- **Log-MinMax:** Logarithmic transformation to compress peaks.

Model Architectures

Machine Learning:

- Random Forest, AdaBoost, XGBoost

Deep Learning:

- MLP, CNN, Transformer